

Instructions:

- **Due:** March 2, 11:59PM . Upload one **single PDF** to Blackboard containing your complete submission and upload all associated L^AT_EX source files.
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page.
- Label your PDF file exactly as

ENME408_S26_HW04_YourLastName.pdf

- All work must be typeset in L^AT_EX. Handwritten or scanned submissions will not be accepted.
- Failure to follow these submission and formatting guidelines may result in a substantial point deduction or a score of zero.

Problem 1. Consider a solid cube $\mathcal{C}$ of side length ℓ and mass m . Consider a frame F_A whose axes are aligned with the edges of the cuboid. Let the center of the cube be denoted by c . The moment of inertia of $\mathcal{C}$ with respect to c resolved in F_A is

$$\left. \vec{J}_{\mathcal{C}/c} \right|_A = \frac{1}{6} m \ell^2 I_3. \quad (1)$$

1. Consider a frame F_B whose axes are aligned with the diagonals between the opposite corners of the cube. Compute $\left. \vec{J}_{\mathcal{C}/c} \right|_B$.

Hint: Recall that $\left. \vec{M} \right|_A = \mathcal{O}_{A/B} \left. \vec{M} \right|_B \mathcal{O}_{B/A}$.

2. Consider the corner $o = [0 \ 0 \ 0]^T$. Note that $c = [\frac{1}{2} \ \frac{1}{2} \ \frac{1}{2}]^T$. Compute the moment of inertia of $\mathcal{C}$ with respect to o resolved in F_A , that is, $\left. \vec{J}_{\mathcal{C}/o} \right|_A$.

Hint: Use the parallel axis theorem $\vec{J}_{\mathcal{B}/z} \triangleq \vec{J}_{\mathcal{B}/c} + m_{\mathcal{B}} \left(|\vec{r}_{z/c}|^2 \vec{I} - \vec{r}_{z/c} \vec{r}_{z/c}' \right)$.

Problem 2. Consider the link $\mathcal{L}$ connected to the ceiling $\mathcal{C}$ with a weld joint at w shown in Figure 1. The external force $\vec{f}_y$ is applied to $\mathcal{L}$ at y . Assume that the distance between y and w is ℓ . Assuming that $\mathcal{L}$ is in static equilibrium, find the reaction force and the reaction torque applied to the link $\mathcal{L}$ at w .

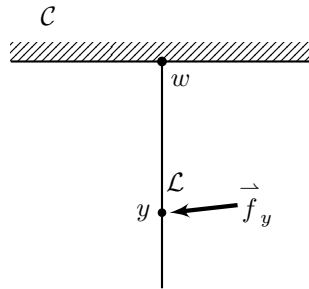


Figure 1: Link $\mathcal{L}$ and ceiling $\mathcal{C}$ connected by a weld joint at w . $\mathcal{L}$ applies a reaction force to $\mathcal{C}$ and a reaction torque to $\mathcal{C}$ balanced at w .

Problem 3. Consider a planar bicopter operating in the vertical plane. The vehicle consists of a rigid body of mass m with two identical rotors located at a fixed distance ℓ from the center of mass, as shown in Figure 2. The propellers at a and b apply positive forces f_1 and f_2 in the opposite direction of $\hat{k}_B$, respectively. Gravity acts along $\hat{k}_A$.

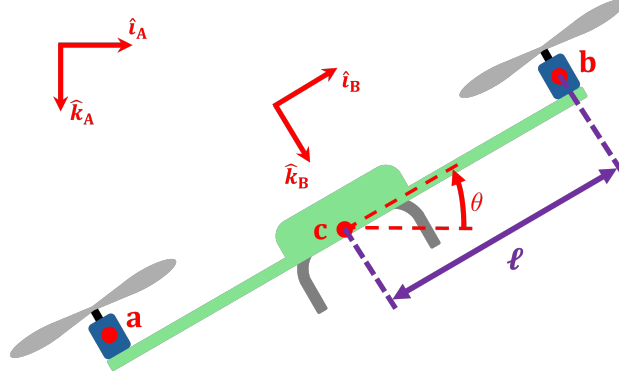


Figure 2: Diagram of a bicopter.

1. Draw a free-body diagram (FBD) of the bicopter and label all forces. Indicate the body-fixed frame and inertial frame on your sketch. Express the forces in both the body-fixed frame and the inertial frame.
2. Compute the net moment about the center of mass due to all forces.
3. Define a point w fixed on ground and define $\vec{r}_{c/w} = r_1 \hat{i}_A + r_2 \hat{k}_A$. Using Newton's second law, write the translational equations of motion.
4. Denote the physical moment of inertia matrix of the bicopter by $\vec{J}_{B/c} = J_2 \hat{j}_A \hat{j}_A'$. Note that the other components of the physical moment of inertia matrix are ignored since the rotation is constrained around the $\hat{j}_A$ axis. Using Euler's equation, write the rotational equations of motion.